

Muhammad Junaid Farooq

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Software Engineer | Full-Stack Development & AI/ML

SUMMARY

Software Engineering graduate and AI/ML Trainee at NETSOL Institute of Artificial Intelligence. I build and deploy full-stack applications with React, Next.js, Node.js, Express.js and MongoDB, including real-time features, Stripe payments, secure authentication and automated API tests. I also built a reproducible machine learning pipeline that compares three regression models (selected Random Forest: R^2 0.898 on held-out data) and serves predictions through a tested FastAPI app.

EDUCATION

The Islamia University of Bahawalpur (IUB)

Bahawalpur, Pakistan

Bachelor of Science (BS) in Software Engineering

July 2026

- CGPA: 3.8 / 4.0
- **Related Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Engineering

Punjab Group of Colleges (PGC)

Khanpur, Pakistan

Faculty of Science (Pre-Engineering)

June 2022

SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL

AI/ML & Data: Scikit-learn, Pandas, NumPy, Matplotlib, EDA, Data Preprocessing, Regression, Cross-Validation, Model Evaluation

Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, Socket.IO, JWT, Auth.js

Frontend: React.js, Next.js, Redux Toolkit, React Query, Tailwind CSS

Databases: MongoDB, Mongoose, MySQL, Schema Design, Indexing

Tools, Deployment & Integrations: pytest, Supertest, Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Vercel, Render, MongoDB Atlas, Stripe, Cloudinary, Mapbox

Currently Learning: Deep Learning, PyTorch, Computer Vision, Speech Recognition

EXPERIENCE

NETSOL Technologies Inc.

Lahore, Pakistan

AI/ML Trainee, NETSOL Institute of Artificial Intelligence (NIAI)

Sep 2026 – Present

- Completed coursework in mathematics and statistics for ML, data preprocessing, supervised learning and model evaluation; currently covering deep learning, unsupervised learning and MLOps.

Arch Technologies

Remote

Web Development Internship

Dec 2025 – Jan 2026

- Built the REST API (Node.js, Express.js, MongoDB) and Next.js frontend of a professional social networking platform, covering authentication, user profiles, connection requests and posts.
- Implemented token-based authentication, RESTful APIs and Redux Toolkit state, using Git feature branches and pull requests.

PROJECTS

Medical Insurance Cost Predictor

Machine Learning Project

Sep 2026

- Built a reproducible scikit-learn pipeline on 1,337 US medical insurance records: duplicate removal, a fixed 80/20 split, and one-hot encoding inside the pipeline to prevent data leakage.
- Compared Linear Regression, Random Forest and Gradient Boosting with 5-fold cross-validation; the selected Random Forest reached R^2 0.898 and MAE \$2,445 on held-out test data (linear baseline: R^2 0.807, MAE \$4,177) and eliminated negative predictions.
- Served the model through a FastAPI REST API and web form with training-range input validation, permutation-importance insights and 10 pytest tests; deployed on Render.

Tech: Python, Pandas, NumPy, Scikit-learn, FastAPI, Pydantic, pytest, Render

Live: medical-insurance-charge-predictor.onrender.com

GitHub: github.com/muhammadjunaidfarooq/medical--insurance-charge-predictor

OmniMart – Grocery Delivery Platform

Full-Stack Project

June 2026

- Built a full-stack grocery delivery platform (Next.js, TypeScript, MongoDB) for 3 user roles (customer, admin, rider) with 27 REST API endpoints and 5 MongoDB models.
- Added a separate Node.js/Express + Socket.IO server with 8 real-time events for live order updates, geospatial rider dispatch (2dsphere index, nearest free riders within 10 km), GPS tracking and per-order chat.
- Implemented Auth.js login with role-based access, server-side pricing, Stripe Checkout with webhook verification, and race-safe job acceptance using atomic conditional updates.

Tech: Next.js, TypeScript, MongoDB, Socket.IO, Express.js, Auth.js, Redux Toolkit, Stripe, Cloudinary, Leaflet, Vercel, Render

Live: grocery-delivery-app-omega-lilac.vercel.app

GitHub: github.com/muhammadjunaidfarooq/grocery-delivery-app

Professional Network – LinkedIn-Style Social Platform

Full-Stack Project

Jan 2026

- Built a LinkedIn-style platform (Next.js, Redux Toolkit, Node.js, Express.js, MongoDB) with profiles, people search, connection requests and a paginated feed with likes and comments.
- Designed 27 REST API endpoints with ownership checks and input validation; secured authentication with bcrypt, hashed expiring session tokens, rate limiting, Helmet and strict CORS.
- Wrote 29 API integration tests (Node.js test runner + Supertest); added image uploads (Multer + sharp, converted to WebP) and PDF resume export; deployed on Vercel, Render and MongoDB Atlas.

Tech: Next.js, React, Redux Toolkit, Node.js, Express.js, MongoDB, Mongoose, sharp, PDFKit, Vercel, Render

Live: fullstack-professional-network.vercel.app

GitHub: github.com/muhammadjunaidfarooq/fullstack-professional-network

WanderLust – Geospatial Rental Platform

Full-Stack Project

Sep 2025

- Built an Airbnb-style rental marketplace (Node.js, Express.js, MongoDB, EJS, MVC) with 14 RESTful routes, Mapbox geocoding with interactive maps, and Cloudinary image uploads.
- Added Passport.js authentication, owner-only authorization middleware, Joi validation and centralized error handling.

Tech: Node.js, Express.js, MongoDB, REST APIs, Mapbox, Cloudinary

Live: online-rental-marketplace-geospatial.onrender.com/listings

GitHub: github.com/muhammadjunaidfarooq/wanderlust-geospatial-rental-platform

Khamosh – Offline Urdu/English PII Redaction Tool

AI / Speech Project

In Progress • Oct 2026

- Building an offline tool that detects spoken CNIC and mobile numbers in Urdu-English audio/video (faster-whisper speech-to-text, FFmpeg/pydub audio processing) and mutes or beeps them.

CERTIFICATIONS

Udemy

100 Days of Code: The Complete Python Pro Bootcamp | [View Certificate](#)

Feb 2024

Apna College

Sigma 6.0 – Full Stack Web Development | [View Certificate](#)

Oct 2025

Aspire Institute

Aspire Leaders Program | [View Certificate](#)

Oct 2025